

A secured novel classifier for effective classification using Light Gradient Boosting Machine(LightGBM) Algorithm

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Abstract— This paper presents an anomaly detection method based on LightGBM(Gradient Boosting Machine) classification methodology. To improve the privacy and security of the data in the cloud environment, the RSA algorithm is used. For providing security, the RSA algorithm is employed for the key generation process. Furthermore, to perform the classification task, the LightGBM algorithm is used within the cloud environment to classify the anomalies. For efficient classification, feature selection is an important task. So, RFE(Recursive Feature Elimination) algorithm is employed to select the best features. To evaluate the proposed method, we conducted experiments on the NSL-KDD dataset and compared the results with the Decision Tree algorithm. Experimental results have shown that the proposed model obtained better performance than the Decision Tree classifier in terms of classification accuracy.

Keywords— Light Gradient Boosting Machine, Anomaly, classification, Recursive Feature Elimination, Decision Tree

I. INTRODUCTION

In today's computing world, one of the fastest, flexible and most used technologies in the IT (Information Technology) sector is cloud computing technology [1]. Cloud computing offers various on-demand services to users. The usage cost is based on the services accessed by the user, which is named pay-as-usage [2]. The cost of accessing these services from the cloud is less. Cloud computing includes network, storage and other computing resources which are combined as a pool of cloud resources. It allows many users to share resources, which increases the usage of resources [3]. Cloud provides platform services to develop various software, Infrastructure services to access hardware resources and software services to develop various applications by the users. The recent survey by Gartner stated that the use of cloud computing in the industries is increasing at a rate of 40%. It is predicted that the rate increases by greater than 25% per year, with a total income of \$55 billion [4]. Storage services are considered important services since most of the users migrate their huge volume of data from their local system to the cloud systems.

Moreover, when performing computations at the local system, it is not capable of handling such a massive amount of data and the computation time required is more. So, to make the computation in a fast and smooth manner, users are started using cloud services. Though the cloud is offering all these valuable services, the security in the cloud remains a challenge in accessing the cloud services.

In the cloud, the two security issues to be focused on are; first, one focus on the service provider should assure the cloud services delivered, and the infrastructure is secure and safe, the second one guarantee the users that the data and information of the users are secured[6,7]. These security challenges threaten the users to store their private data in the cloud. Data confidentiality, data integrity and privacy preservation are considered the biggest hurdles against the adoption of cloud storage [8]. Thus, data privacy and security issues make the researchers develop various cryptographic algorithms. Cryptographic algorithms increase the security and privacy of the data by converting the raw or original data into an unreadable format known as ciphertext. There are two main types of cryptographic algorithms, such as Symmetric key cryptography and asymmetric key cryptography. In symmetric-key cryptography, a single key is shared among all the authorized users. This single key is known as the secret key, which is shared among the users. The shared secret key is used for both the encryption and decryption process. The encryption process is used to convert the data into some other format known as the ciphertext to store in cloud storage. The decryption process is used to convert cipher text to plain text while retrieving it from the cloud data storage. Since a single secret key is shared among several users, security is again a major problem, since an unauthorized user can hack the key easily to access the original data. Hence, asymmetric key cryptography or public key cryptography came into existence. In this scheme, there are two keys, known as public and private keys. The encryption process uses the public key, and the decryption process uses the private key. Here, the private key is the secret key. The user who is having the private key alone can decrypt and view the original data. The security of the data is greatly increased with this technique. Asymmetric cryptography technique is thus more powerful than symmetric key cryptography. Among the public key cryptographic algorithms, the RSA algorithm is the most powerful algorithm since the key size generated is more, so it is not easy to break the key. Hence, the RSA algorithm is used in the proposed approach for privacy preservation and security of the data stored in the cloud.

Data anomaly is the abnormal behaviour of the data. There are various approaches to detect the abnormalities present in the data. Today machine learning algorithms are employed to use in anomaly detection systems [5]. Both the types of machine learning algorithms namely supervised and unsupervised methods are used in

the anomaly detection process. Both these techniques use classification and clustering approaches respectively. In some applications, semi-supervised algorithms are also used. LightGBM algorithm is used in our work, for classifying the anomalies. While processing large datasets to detect anomalies, feature selection is an important task to select the required features among a large number of available features. Significant feature selection plays a vital role in improving the accuracy of the anomaly detection model. We have used the RFE algorithm along with logistic regression to select the important and required features. The processing time greatly varies based on the number of chosen features. Varying the features may vary the classification accuracy also. The standard dataset is always used for the experimental validation of the proposed model. To obtain accurate validation results, the dataset plays an important role. So, the NSL-KDD dataset in our work, which is a standard dataset used for intrusion detection purpose.

The remaining portions of the paper are structured as follows: the related work on cloud based anomaly detection using machine learning is discussed in section 2. The system model of the proposed work is presented in section 3. Section 4 presented the privacy preservation and security of the data using the RSA algorithm. Feature selection using the RFE algorithm is discussed in section 5. Anomaly detection using the LightGBM algorithm is discussed in section 6. Section 7 presented the Results and Discussions. Finally, Section 8 presented the conclusion and the future work.

II. RELATED WORK

A. Selecting a Template (Heading 2)

In [9], the authors combined the two supervised machine learning algorithms namely Naive Bayesian classifier and Iterative Dichotomiser 3 (ID3) and proposed a hybrid model to analyze the network data and the complex properties of network attacks. The detection speed and accuracy is improved. Huang et al.[10] stated that the number of observed features is reduced into a smaller set, which is sufficient to obtain the information necessary to the goals of the anomaly detection technique. Feature selection methods are used to reduce the dimensionality of the datasets. Thottan et al.[11] reviewed the anomaly detection methods which used artificial intelligence, machine learning and state machine modelling. Garg et al.[12] proposed an Ensemble-based classification model for network anomaly detection in massive datasets and stated that data with a large number of features cause overfitting problem and the overall performance of the model get reduced. Thus optimal feature selection is required to reduce the computational overhead and over-fitting problems. Xu et al. [13] proposed a dynamic extreme learning machine for the classification of patterns present in the continuous data stream. Their proposed technique was faster, but the accuracy rate was less. Moshtaghi et al. [14] proposed a model for anomaly detection in data streams and fuzzy rule-based methods were used to learn the incoming samples. Columbus[15] stated that 49% of businesses are deferring cloud utilization due to cybersecurity problems. Hence, privacy and security are important factors to be considered in cloud technology. Recently, anomaly

detection emerged as the main research area among researchers [16,17,18,19].

To summarize, the advancements in the cloud and machine learning techniques provide a way for developing a robust model for anomaly detection techniques. We have followed these technological advancements to develop a robust, reliable and flexible anomaly detection model.

III. SYSTEM MODEL

The proposed system model consists of the components such as user, Trusted Authority and Computation engine. Among these, the Trusted Authority and Computation engine components reside in the cloud environment. The main components in the system model in fig.1 are, (1) Trusted Authority (TA) and (2) Computation Engine(CE). The main functions of the TA are: (i) Validating the user, (ii) Issuing public key for encryption, (iii) Performing decryption and (iv) Providing classification result back to the user. The role of the CE are (i) Feature Selection, (ii) Classification (Anomaly detection) and (iii) Return Analysis Report.

The cloud environment consists of the TA and the CE. The user sends a request for the public key to the job. The public key is used to encrypt the data for the secure transmission of data to the cloud. The TA generates the key pair(Pp, Pr) for a job, where, Pp and Pr are Public keys and Private key respectively. These keys are used for encryption and decryption purposes. TA validates the user by using the credentials supplied by the user and provide the public key(Pp) to the user. The credentials are the username and the password. The user encrypts the data with a Pp key and sends the encrypted data to the TA. The user submits a ticket for Anomaly Detection to the TA. TA initiates the CE with the user request. TA gets the encrypted data from the user and decrypts it using the Pr key. After decryption, the data is provided to the CE where the optimal number of features are efficiently selected by the RFE algorithm to improve the classification accuracy. Thus the computation overhead of the system is greatly reduced. Upon selection, the anomaly detection is performed by using the LightGBM classification Algorithm. After completing the classification task, the final result is sent to the legitimate user from the computation engine through the Trusted Authority.

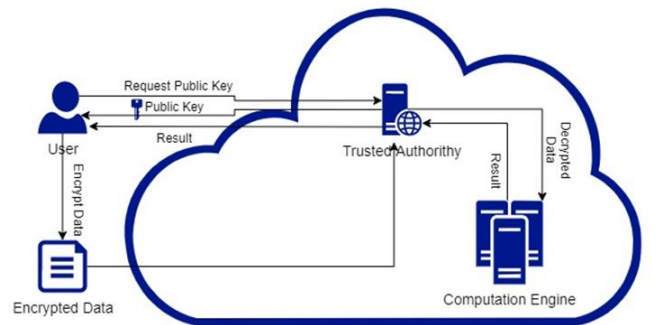


Fig 1. System Model

IV. PRIVACY PRESERVATION AND SECURITY USING THE RSA ALGORITHM

Our proposed cloud based privacy preserving model adopting the RSA scheme can provide end-to-end privacy and confidentiality. In RSA key generation, a public key is

an integer value generated randomly by using the two prime numbers(p and q). With the help of the public key, the original data is well-encrypted. Algorithm 1 describes RSA key generation. The private key is used for decryption. Both the keys were generated by the TA.

Algorithm 1: RSA algorithm for key generation

Begin

Choose two large prime numbers p and q

$$n = p \times q$$

$$\phi(n) = (p-1) \cdot (q-1)$$

e= random value, such that $1 < e < \phi(n)$

Determine $d = e^{-1} \bmod \phi(n)$

Public key = (e, n) i.e. e is a public key;

private key = (d) ;

Stop

V. FEATURE SELECTION

Feature selection [20,21] has become essential to choose the finest and optimal features and applied in several fields such as the medical field, biology, economics, manufacturing, image processing and production. Feature selection plays a vital role in the classification task. It reduces computational time and increases classification accuracy. Recursive feature elimination (RFE) is a feature selection algorithm that selects the best features and eliminates the worst features based on the ranking of features. For ranking logistic regression method is used. RFE with logistic regression evaluates the features and select the best features by using the ranking score. The ranking of features is calculated by Logistic Regression using the eqn. 1:

$$(P(Y))/(1+P(Y)) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n \quad (1)$$

where, Y = Target (Dependent variable)

$x_1, x_2, \dots, x_n$ = Features (Independent variables)

β_0 = Constant value

$\beta_1, \dots, \beta_n$ = Co-efficient values

The probability score value or the rank for each feature is calculated by using Y and the features $x_1, x_2, \dots, x_n$ as inputs. It can be calculated using the eqn.2

$$P = 1 / (1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n)}) \quad (2)$$

where P=Probability value or Rank value

Thus the rank value for every feature is assigned. The total number of features in the dataset are denoted as 'N' in fig 2. Feature Set (FS) contains all the features and Feature Rank (FR) set contains the features with rank values. The feature with the highest rank is selected as the best feature and the lowest ranking features were eliminated. The selected features provided to LightGBM for the anomaly detection process.

VI. ANOMALY DETECTION USING LIGHTGBM ALGORITHM

Light BGM [22] is a gradient boosting framework, and it is a powerful algorithm. It uses a tree based learning methodology. The computation speed is very fast for this algorithm so that it has the name light. It can able to process a large volume of data, and it requires less amount of memory. The LightGBM performs classification by using the selected features. Algorithm2 shows the LightGBM algorithm

The following are the main parameters [23,24,25] of the LightGBM which are used to implement the proposed model:

number_leaves – the leaves present in each tree.

learn_rate – learning rate of the algorithm.

maximum_depth – algorithm's maximum depth.

boosting type – it defines the type of the algorithm

minimum_data – the minimal amount of data in the leaf node

feature_fraction – the value ranges from 0 to 1. It is the ratio of the particular feature to the total number of features.

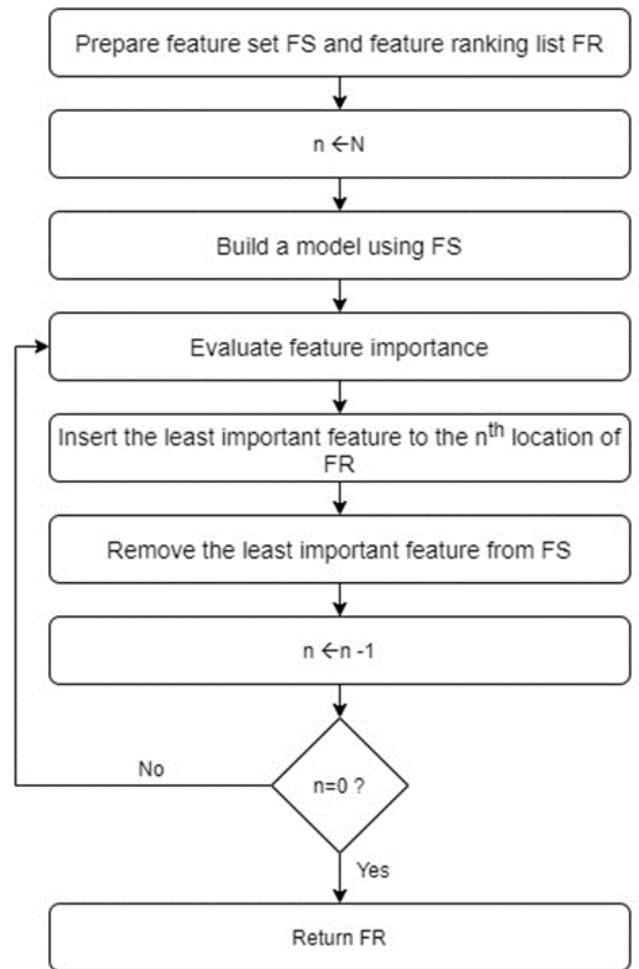


Fig. 2: Feature selection using RFE algorithm.

The parameter values for our implementation is set as follows:

```

learn_rate = 0.003
booster_types = 'gbdt'
objective_fn = 'binary'
metrics = 'binary_logloss'
sub_features = 0.5
number_leaves = 70
minimum_data = 50
maximum_depth = 70

```

Algorithm 2: LightGBM

```

Input: T= training data
Input: F= iterations
Input: u= sampling ratio of large gradient data
Input: v= sampling ratio of small gradient data
Input: loss_fn= loss function, L_weak= weak learner
Model ← {}, factr ← (1-u)/v
topn ← u × len(T), randn ← v × len(T)
for i = 1 to F do
    pred ← Model.predict(T)
    k ← loss_fn(T, pred)
    w ← {1, 1, ... }
    sort ← GetSortedIndices(abs(k))
    topset ← sorted[1:topn]
    randset ← RandomPick(sorted[topn:len(T)], randn)
    usedset ← topset + randset
    z[randset] = factr
    newml ← L_weak(T[usedset], -
k[usedset], z[usedset])
    Model.append(newml)

```

VII. RESULTS AND DISCUSSIONS

For experimental validation, the standard NSL-KDD dataset is used. It contains 42 features or attributes. Among these, only the important features are selected by using the RFE algorithm. For classification by LightGBM, these selected features are used by it. The performance of LightGBM is compared with the Decision Tree algorithm, and the accuracy rate of LightGBM is more than that of the Decision Tree. The metrics used for evaluation are:

- i. Accuracy
- ii. Precision and
- iii. Recall

Accuracy

Accuracy is the measure to calculate the performance of the model. It is the ratio of correctly predicted values to the total values. The accuracy is calculated by using the eqn. 3.

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} \quad (3)$$

where, TP is the Total True Positive values, TN is the Total True Negative values, FP is the Total False Positive values and FN is the Total False Negative values.

The classification accuracy of LightGBM is obtained as 0.99836, and the classification accuracy of Decision Tree is obtained as 0.986 as in fig 3.

The analysis is performed by varying the count of features selected. The overall features obtained after performing the Label and one-hot encoding schemes is 122. We performed the analysis by choosing the features ranging from 10 to 90. Based on the analysis, we found that the performance observed is very low when the number of features was very low, that is, in our case, low indicates 10 features. The best accuracy was observed when the number of features was 50. It was also observed that after the idle range of 50, the performance was not increasing and it was stable at around 0.998.

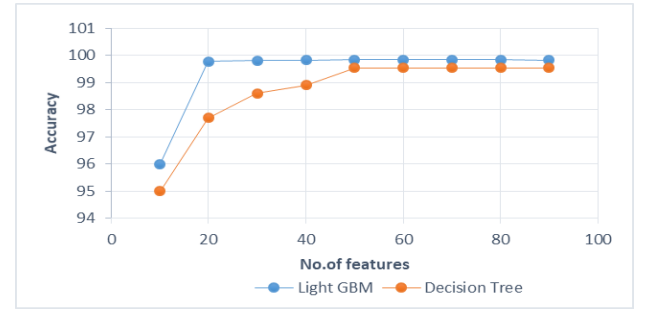


Fig.3: Accuracy value of LightGBM and Decision Tree by varying the number of features

Precision

Precision is the measure of the ratio between true positive predictions obtained and the total positive predictions available. It is calculated by using the eqn.4:

$$\text{Precision} = \frac{TP}{TP+FP} \quad (4)$$

The precision values obtained for Light GBM and Decision tree classifier in the proposed model are in fig 4. The precision value is calculated for both the algorithms by varying the number of features. It shows that Light GBM obtained more precision value than Decision Tree classifier. When the number of selected features were more than 50, then the precision value obtained for Light GBM is 0.998 and for Decision Tree is 0.992 and remains the same for the remaining number of features.

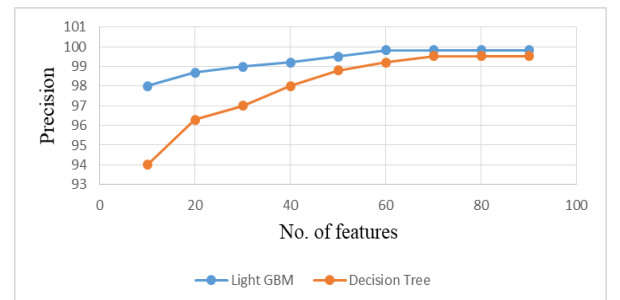


Fig. 4 Precision value of LightGBM and Decision Tree by varying the number of features

Recall

The recall is the measure of the ratio between true positive and the sum of true positive and false negative values. It is measured by using eqn.5:

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN}) \quad (5)$$

The recall values obtained for Light GBM and Decision tree classifier in the proposed model are in fig 5. The recall value is calculated for both the algorithms by varying the number of features. It shows that Light GBM obtained more recall value than the Decision Tree classifier. When the number of selected features were more than 50, then the recall value obtained for Light GBM is 0.992 and for Decision Tree is 0.98 and remains the same for the remaining number of features.

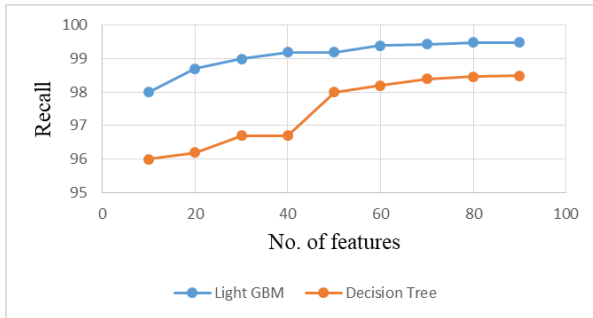


Fig. 5 Recall value of LightGBM and Decision Tree by varying the number of features

We also compared the performance of the LightGBM algorithm with the Decision Tree, and we could found that the performance of LightGBM is slightly higher than the Decision Tree. The same number of features are used in the Decision tree as that of LightGBM. The accuracy of the Decision Tree was observed as 99.53 while the performance of LightGBM was 99.83 as in fig.6. Hence the performance of LightGBM is more than that of the Decision Tree classifier.

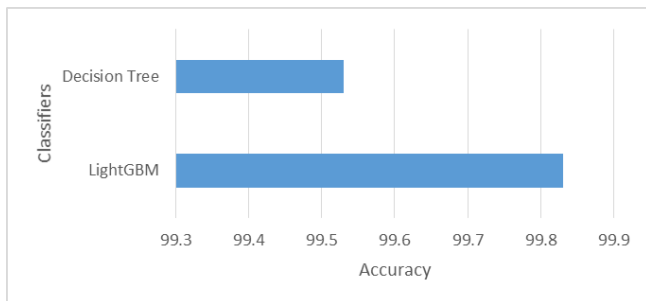


Fig.6 Comparison of LightGBM with Decision Tree classifier

VIII. CONCLUSION

In this paper, an anomaly detection model based on the cloud environment is proposed by using the machine learning algorithm LightGBM for classification and RFE for feature selection. Moreover, the RSA algorithm is used for privacy preservation and security of data in the cloud.

The Trusted Authority in the cloud environment is responsible for issuing the keys to the user and the computation engine. The computation engine is responsible for performing feature selection and classification tasks. The experimental validation is performed on the standard NSL-KDD dataset to analyze the performance of our work on the anomaly detection process. The performance of the LightGBM algorithm is compared with the Decision Tree algorithm with the same dataset. The proposed model provided high accuracy than the Decision Tree in detecting data anomalies.

In future, the proposed model would be evaluated with real-time datasets to work for real-time data. Also, the performance of the proposed model would be compared with various other machine learning algorithms.

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